

PrivateAIDELPHI: Adopting and adapting private AI for risk assessment of safety critical systems

Nikolaos Papakonstantinou¹, Douglas Van Bossuyt², Ryan Bell², Ryan Longshore² and Mika Heikkilä¹

¹Fortum Power and Heat Oy, Espoo, Finland

²Naval Postgraduate School, Monterey CA, USA

Key Words: Risk assessment, Private AI, Delphi, Large Language Model, Safety, Security, Resilience

SUMMARY & CONCLUSIONS

Recent advancements in generative AI and Large Language Models (LLMs) have produced tools able to convincingly answer questions and instruct humans through complex tasks. This is achieved after a computationally intensive training based on a large set of documents. Although the results can be impressive, there are cases of problematic responses. Even though there is significant motivation and interest to introduce these kind of AI tools in safety engineering processes, it is recognized that this needs to be done carefully and with human experts in control. Additionally, it is common that system design information is sensitive and cannot be shared with cloud-based AI platforms.

The PrivateAIDELPHI method presented in this paper introduces “AI experts” in early risk identification and scoring for complex safety critical systems. It extends the established Delphi iterative method for reducing subjectivity and reaching consensus among experts. These AI experts are enabled by tools that can be hosted locally and without a connection to the Internet.

The benefit of introducing AI experts, with the current technology is the potential to get different perspectives and trigger/inspire human experts to expand their view.

The basic steps of the PrivateAIDELPHI are the definition of the questions to be answered, the preparation of the background information material and then the collaboration of human and AI experts on the identification and scoring of risks. The method includes sanity checks by humans on the AI-generated answers as well as the fact that the final risk prioritization step involves only humans. Practically the AI with its answers is used to expand the thought horizon of the human experts and not to just provide an answer.

Although the PrivateAIDELPHI is designed to operate using confidential data (system design and background information), for practical reasons it was demonstrated in this paper using public information on the NuScale Small Modular Reactor (SMR) design and a small set of relevant public documents giving background on nuclear system risks. Four AI models (three that can be considered private/local and ChatGPT) were used, this case study did not experiment on the

optimization of the LLMs and the results are not to be seen as a comparison/benchmark of the different AI models (between them or against humans).

The integration of human and AI experts in the limited NuScale case study showed that the AI can bring valuable input to the group of experts tasked with risk identification and prioritization, thus it was successful. While not being able to express risks in a systematic manner (the AI results can be seen more like talking points - food for thought) the integration of AI experts together with humans does enhance the risk assessment process. It is very important to highlight that humans are in control and finally responsible for the final results.

1 INTRODUCTION

Risk assessment is a key element of resilience engineering for complex systems. The basic aspects of risk assessment during early design are the identification and prioritization of safety risks as well as security threats based on intelligence covering the environment of the system under study, experience from past projects as well as any available early design information. For complex sociotechnical cyber-physical systems this process stretches the experience and forecasting capability of practitioners involved and it is perceived as a very difficult and the results are highly subjective depending on the background, experience and interests of the practitioners. The risks and threats identified/prioritized during the early design phase have a great impact to the system development process and mistakes can lead to costly redesign or reduced system resilience. A more objective and scalable approach to risk assessment for complex systems is needed.

The CyberRiskDELPHI method [1] developed in past work showed that the traditional Delphi method [2] for reducing subjectivity can be integrated to state-of-the-art risk/threat assessment methods and help to improve the results by using groups of experts with diverse experience and background. It uses an iterative approach with multiple rounds of collecting identified risk and threats as well as scoring them. This CyberRiskDELPHI approach worked well, but still it was heavily impacted by the selection/availability of experts. It is not easy to build a multi person team with the needed diversity

and available time to go through multiple risk assessment iterations. In that past work the use of cloud based large language model was suggested (and briefly tested) as a future direction to help mitigate some of the shortcomings of the proposed method, but this approach had the practical challenge that system design information is usually confidential and not possible to be shared with AI systems not locally hosted and private. The AI at that time gave very generic results of little use because it didn't have access to the case study information, but it was evident that there was great potential, if private AI could be utilized which would be able to operate on confidential data.

The scientific contribution of this paper is the extension of the Delphi method with the use of AI large language models which can be hosted privately (local computation and data storage, without the need for a connection to the Internet during operation). These private "AI experts" can use confidential information and their results can be more relevant to the system under study. In the case study a non-private AI platform was also used (ChatGPT), since the data was not sensitive. Special focus is given to the traceability of the replies of these AI participants to the domain and system specific data sources given to them.

The proposed PrivateAIDELPHI method is demonstrated using 3 AI large language models that are able to be locally hosted and can easily ingest additional information (i.e. without the need to retrain their basic model) and the proprietary ChatGPT. In principle, a combination of AI and human experts participate in this process, in an anonymous way. The Delphi method was extended with a preliminary step focusing on intelligence discovery (domain and system specific information as well as risk assessment guidance) which will result in a set of machine-readable documents that AI and human experts can use. The case study in this paper is from the nuclear energy generation domain and the public information available for NuScale Small Modular Reactor design. For this use case the AI experts were given additional domain specific information as well as design documents in text (PDF) form for the specific system under study.

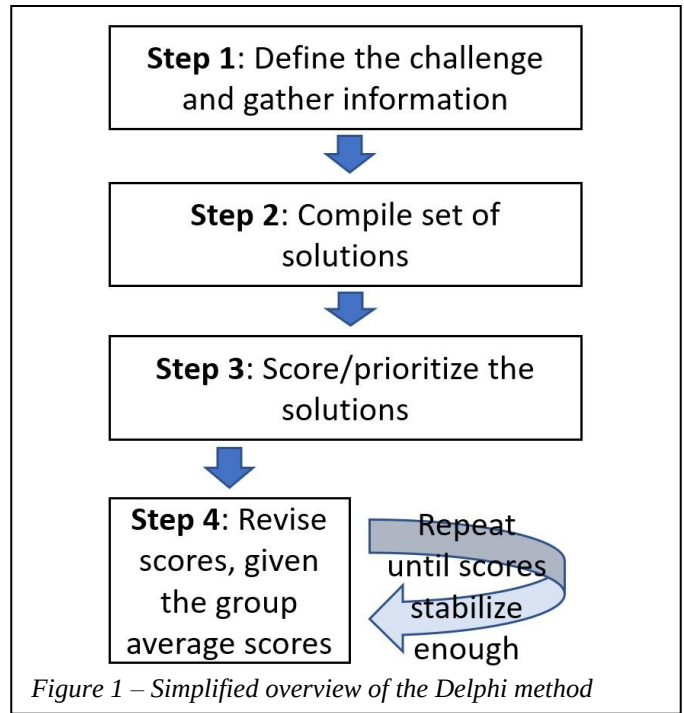
2 LITERATURE REVIEW

This section gives a high-level overview of the risk assessment challenge, the basic principles of the Delphi method and the recent advancements of generative AI and Large Language Models.

2.1 Risk assessment of complex systems

To satisfy the modern societal requirements mission critical infrastructure relies on increasingly complex systems. They are designed to satisfy their system requirements in order to be useful, but also have to follow safety design principles and include support systems to anticipate design basis events as well be resilient against generic risks [3].

Risk assessment, a key part of safety engineering, includes the identification of relevant risks and prioritization of these risks depending on their probability of occurrence and severity of their consequences [4]. A key challenge in risk assessment



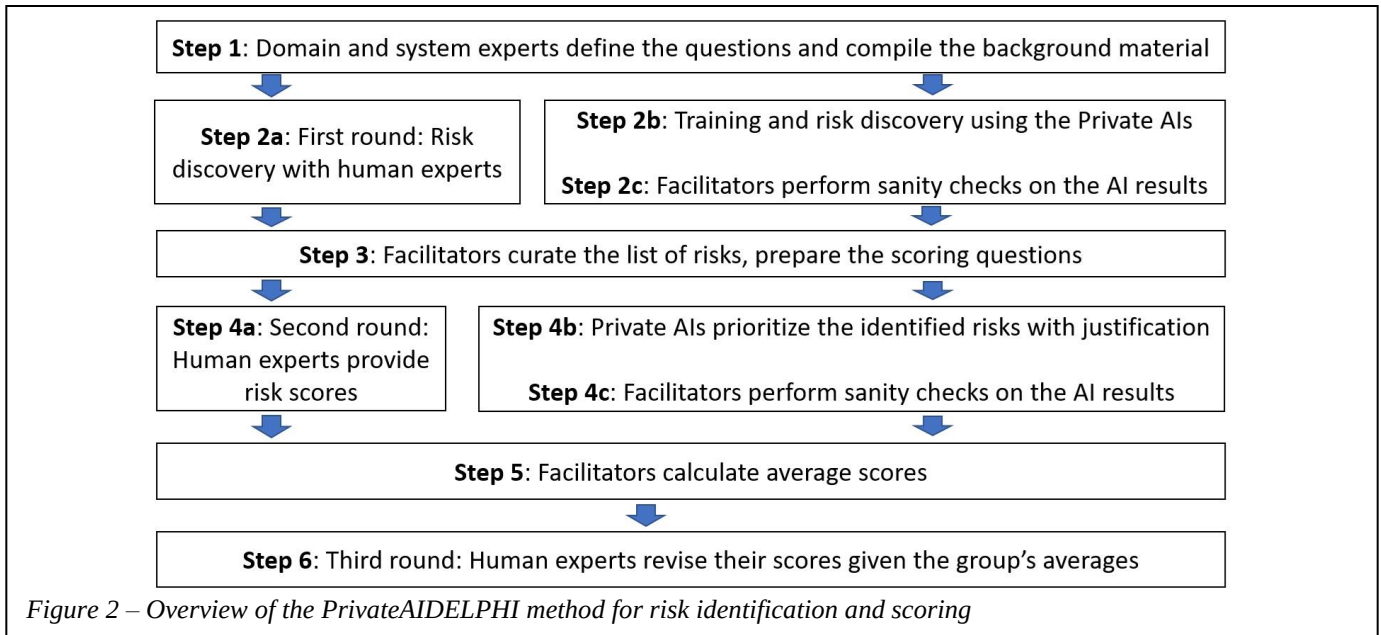
for complex systems include the difficulty in achieving objective and comprehensive risk identification as well as the coverage of the different system aspects. The earlier important risks are identified, the easier it is to introduce mitigations to the system [5, 6]. Early risk assessment, although important, it is often based on incomplete background and system information and thus the resulting estimations are heavily influenced by the background and experience of the practitioners performing the study [7].

2.2 DELPHI method

To address cases where subjectivity and lack of consensus between experts in decision making is problematic the RAND Corporation developed in the '60s an iterative method called Delphi [2]. The goal of the Delphi method is to use a group of experts and an iterative process to reduce the overall subjectivity, increase the coverage of different aspects of the question under study and guide the group towards consensus. A simplified version of the Delphi method is shown in Fig. 1. The first step of the process is for the facilitators to gather a diverse group of experts, define and formulate the questions to be asked and gather related background and system specific information. In Step 2 the experts are asked to provide anonymously quantitative answers to the list of questions (e.g. scores). These answers are averaged and in the most important step of the method the results of the first round of questions are shared with the experts who are asked to revise their score, given the group's average. This iteration can happen more times to improve consensus further, if needed. The final average scores are the results of the method.

2.3 Generative AI – Large Language Models

Large language models (LLMs) are artificial neural



networks trained to perform natural language processing tasks. While Google’s introduction of the transformer architecture enabled rapid LLM advancements, OpenAI’s introduction of a chatbot interface in the form of ChatGPT increased the accessibility of LLMs to the general public and continues to drive public interest in AI technologies [8, 9]. Advances in technology have enabled multi-modal LLMs capable of generating images, audio, and/or video in addition to natural language [10].

LLMs are characterized by their size and quantization. The size of a LLM refers to the number of parameters. For example, a 7B model has seven billion parameters while larger commercial offerings may have trillions of parameters. The quantization of a model refers to the data format of the underlying weights. Models are quantized to reduce the amount of memory required to store and run them [11]. If a LLM’s weights are calculated using 32 bit floating point number during initial training, converting the model to 4 or 8 bit integers will significantly reduce the amount of memory required to store the model. It is up to the user to determine if the quantization of the model is acceptable as there will be some loss of accuracy.

When a user utilizes a chatbot interface to request information from a LLM, the LLM makes inferences based on the user’s text input. LLMs are trained on vast amounts of data, but do not contain domain specific knowledge for many fields. This lack of domain specific knowledge can increase the chance of hallucination, or false outputs, during the inferencing process [12].

There are several methods to provide domain specific knowledge to a model. The simplest method is to provide examples to a LLM. A zero-shot prompt is a request for a LLM to perform a task it has no prior knowledge of. Single, few- and multi-shot prompting is a method to teach LLMs how to perform a task by providing clear example(s) [13]. If problems require multi-step reasoning [14], chain of thought prompting can be incorporated into the examples to show step-by-step

reasoning. When external sources are required to augment the LLM’s knowledge base, retrieval-augmented generation (RAG) may be appropriate. RAG utilizes a database of vectorized supplementary information to provide content from external sources to the LLM [15]. RAG does not change the internal weights of the LLM. Finetuning a model incorporates new information by utilizing the new information to change a small number of weights in the model [16]. Finetuning is not as computationally intensive as training a new model, but is much more computationally expensive and time consuming compared to RAG.

Commercial offerings such as OpenAI’s ChatGPT, Google’s Gemini, and Anthropic’s Claude are available to businesses, academic institutions, and consumers. Commercial models are often large models that require significant computing resources to both train and inference. While these models can perform many tasks well, there are security concerns when injecting sensitive or proprietary data to improve results [17]. The security risk of sharing sensitive or proprietary data outside of an organization is a limiting factor to the widespread use of LLMs at the enterprise level.

To remedy this issue, a user may choose to deploy a local (private) instance of an open source LLM. Local LLMs can be disconnected fully from the internet or hosted in private and secure cloud environments. Low-/no- code tools such as h2oGPT [18], LM Studio, GPT4All, and AnythingLLM [19] enable users to download and run open-source models from LLM hosting sites while streamlining the process for adding context to a model through RAG. Users download a LLM from a hosting site, such as HuggingFace, with the appropriate number of parameters and quantization based on their available computing resources. LLaMa, Falcon, and Mistral are three common open source LLMs available in a variety of sizes and quantizations, but there are many more available on LLM model sharing sites with new LLMs released daily. Personal computers with consumer grade graphics processing units

Table 1 – Summary of the AI expert answers when asked to list the 5 more important risks for the NuScale design.

AI expert 1	AI expert 2	AI expert 3	AI expert 4
1) Unproven Safety of Compact Designs 2) Relaxation of Existing Safety Regulations 3) Lack of Comprehensive Independent Reviews 4) Design-Basis Accidents 5) Operational Safety	1) Loss of Cooling 2) Radiation Exposure 3) Accidents and Meltdowns 4) Containment Failure 5) Operational Safety and Human Error	1) Loss of Coolant Accidents (LOCA) 2) Human Error and Operational Failures 3) External Hazards (Natural and Man-Made) 4) Spent Fuel Pool Safety 5) Passive Safety System Reliability	1) Insufficient Containment Strength 2) Inadequate Emergency Planning Zones and Operator Staffing 3) Inadequate Design-Basis Accident Protection 4) Higher Risk of Nuclear Accident due to Increased Number of Reactors 5) Insufficient Regulatory Oversight and Safety Standards

(GPUs), ample memory and storage are capable of inference on smaller LLMs at higher quantization. Inferencing or finetuning LLMs with a large number of parameters and/or low quantization often require more dedicated computing resources. In these scenarios, dedicated computing or a secure cloud environment may be more appropriate.

3 METHODOLOGY

The PrivateAIDELPHI method extends the Delphi method by introducing “AI experts” and customizes it for the task of early risk identification and scoring for complex systems. The main principles are that AI experts join the expert group after a sanity check of their replies and that the final results are produced only by human experts, who have the overall control of the process.

An overview of the methodology is shown in Fig. 2. It will be presented in this section in a generic way and then in more detail over the example in the case study section.

In Step 1 the facilitators, together with experts, define the questions to be addressed and gather an information package covering the system domain (e.g. defense, nuclear, healthcare) specific as well as what is known about the system under study (e.g. requirements, functional description, design overview) which will be shared with a group of human and AI experts to help them answer the questions on risk identification and prioritization.

In Step 2a the human experts are asked to identify risks relevant to the system and in parallel in Steps 2b the AI LLMs represented as “AI experts” are asked to do the same but with justification and references to sources. In Step 2c the answers of the AI Experts go through a sanity check by the facilitators based on the justification provided and a cross-check with the sources.

The risks identified in Step 2 are compiled in a set and questions on scoring the importance of the risks are added in Step 3. The questions need to accept answers that are quantitative. In the simplest case an example question can be “Provide a score from 0 to 10 (low to high) for the importance of the risks in the list below” or they can analyze the risk parameters (e.g. occurrence probability, existing prevention and mitigation measures, severity of consequence) and calculate a composite score from these.

In Step 4a the human experts provide the scores for the identified risks in an anonymous way and in Step 4b the AI

experts also provide their answers which are checked in Step 4c.

In Step 5 the scores of human and AI experts are averaged and in the final Step 6 the human experts are asked whether they would revise their scores given the group average. The concept is that an expert who may not be very confident on their score may revise it towards the average while a confident one will not change their score. The scores of Step 6 are averaged and are the final results of the PrivateAIDELPHI risk identification and prioritization method.

4 CASE STUDY

The PrivateAIDELPHI method can accommodate locally hosted and isolated (Private) AI models to enable access to proprietary/confidential background and system information. In this paper the NuScale Small Modular Reactor design was selected because it was possible to have access to public information related to nuclear safety and an extensive general description of the NuScale plant.

Following the methodology (Fig. 2), the set of background information was compiled and consisted of these documents covering some topics on nuclear safety, Small Modular Reactors and the NuScale SMR plant design [20-30]. These documents were used by both human and AI experts.

The expert group consisted of four human experts and four AI LLM models. The AI experts were using these models:

- **AI expert 1:** (h2oGPT frontend) - mistralai/Mixtral-8x7B-Instruct-v0.1
- **AI expert 2:** (h2oGPT frontend) - meta-llama/Meta-Llama-3-70B-Instruct
- **AI expert 3:** ChatGPT/Custom GPT: Nuclear Engineering Expert/ GPT4o (not private)
- **AI expert 4:** (AnythingLLM frontend) AnythingLLM/Groq/llama3-70b-8192/temperature=0.7/AnythingLLM Embedder/LanceDB vector database/similaritythreshold>=0.25/no memory

Although the AI models (apart from the ChatGPT) are open source and can be downloaded and deployed locally, in this paper the online versions were utilized.

The questions to the human and AI experts for risk identification (Step 2) were these:

- **Question (1)** Which are the 5 most important safety risks of Nuclear Energy Plants?
- **Question (2)** Which are the 5 most important safety risks of

Table 2 – Summary of the expert scores when asked to evaluate the risks for the NuScale design. The risks in red were only identified by humans.

Identified risks for the NuScale design	AI 1	AI 2	AI 3	AI 4	Human 1	Human 2	Human 3	Human 4	First round average score	Scores after revision
Unproven Safety of Compact Designs	8	1	7	6	7	4	8	5	5.8	5.5
Relaxation of Existing Safety Regulations	7	1	6	4	9	2	4	5	4.8	5.0
Lack of Comprehensive Independent Reviews	7	1	5	5	5	3	5	5	4.5	4.5
Design-Basis Accidents	6	8	7	8	6	3	4	9	6.4	5.3
Loss of Cooling	9	7	9	9	9	2	4	7	7.0	5.8
Radiation Exposure	8	6	8	7	4	2	5	9	6.1	5.5
Accidents and Meltdowns	10	10	10	10	8	3	6	10	8.4	7.0
Containment Failure	9	9	9	9	9	2	5	8	7.5	6.3
Operational Safety and Human Error	7	5	8	8	9	6	6	8	7.1	7.0
Fire	6	3	6	3	6	3	7	7	5.1	5.0
Flood	5	4	7	2	8	2	8	6	5.3	5.5
Cyber security attack	8	2	7	6	5	5	8	7	6.0	6.5
Terrorist attack	9	2	8	9	9	6	5	9	7.1	7.0

the NuScale SMR Plant design?

The AI experts were also asked to justify why they chose these risks. They provided also references to sources supporting their response. The facilitators used that information for sanity checking the AI replies. An example of the risks identified by the AI experts is shown in Table 1.

The AI results in this case study were very generic and were used as talking points – inspiration for the human experts rather than concrete specific risks. A reason may be the limited detail of the background material as well as a lack of a template which would force the AI experts to structure risks in a systematic way.

When the replies from the experts were gathered, a list of risks was compiled (9 risks from the AI experts and 4 risks only mentioned by the human experts, see Table 2) and the question for the risk prioritization was sent to the experts (Step 3): “Give a score from 0 to 10 to the risks in this list based on the severity of the risk for the NuScale Plant design (from 0 for the least severe up to 10 for the most severe)”.

In Step 4 the scores provided by the human experts and the scores from the AI experts (after sanity checks) were gathered and in Step 5 the average score was calculated (see Table 2 - column “First round average score”). This is the last step where the AI experts participate. The scores and the average are shown in Table 2.

In Step 6 the facilitators ask the human experts whether they would revise their scores given their old score and the group average score (calculated in Step 5). The facilitators gather the new revised scores and average them to produce the final risk scores which is the last column in Table 2.

Noteworthy observations:

- AI models may take into account a randomness parameter (“temperature”) in their replies. When the AI experts 1 and 2 were asked to provide a list of 5 nuclear lists for a second time in the same discussion, the AI expert 1 repeated the same risks (and complained that we asked that before) while the AI expert 2 gave a slightly different list of risks.
- The AI expert 3 (ChatGPT) gave replies which referenced public documents relevant to the final design of the NuScale

plant, including the evaluations from NRC. The other AI experts did not give references to documents other than those provided by the facilitators.

- AI expert 2 gave a score of 1 to the first three risks of the list mentioning that the supplied background information documents didn’t provide enough information to properly score them. There are many ways to handle this, like ignoring the score, accepting the value of “1” or restart the step with additional background information.
- Emphasis should be given to the facilitators’ contribution for sanity checking the AI responses and the overall need for humans to have control and responsibility the process/results.

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BIOGRAPHIES

Nikolaos Papakonstantinou, D.Sc. (Tech.), Docent¹, is an I&C expert at the Finnish nuclear owner operator Fortum Power and Heat Oy. His interests are safety/security risk assessment, system eng. and information interoperability.

Douglas L. Van Bossuyt, PhD², is an associate professor of systems engineering at the Naval Postgraduate School in Monterey, California. His research focuses on the nexus of risk and failure analysis, RAMS, and systems design methods.

Ryan Bell, PE², works at the Naval Information Warfare Center Atlantic (NIWC LANT) and pursues a Ph.D. at in Systems Engineering at the Naval Postgraduate School in Monterey, California.

Ryan Longshore, PE², works at the Naval Information Warfare Center Atlantic (NIWC LANT) and pursues a Ph.D. at in Systems Engineering at the Naval Postgraduate School in Monterey, California.

Mika Heikkilä, M.Sc.¹, is a Group and Portfolio Manager for Instrumentation & Control projects related to nuclear power plants and other facilities as well as line manager at the Finnish nuclear owner operator Fortum Power and Heat Oy.

¹ Fortum, Keilalahdentie 2–4, 02150 Espoo, Finland. Email: name.surname@fortum.com

² NPS, 1 University Circle, Monterey, CA 93943, USA. Email: name.surname@nps.edu